Ovo in Unity 6.6 (Universal 2D)

A step-by-step guide for someone who has never used Unity. Follow the parts in order. The map is REAL geometry in your project (about 1,700 objects you can click, move, delete and copy), not a picture.

0. How the game stays exactly the same

- The web game's level generator is recorded shape by shape (every filled polygon and every thick line, in drawing order). Those shapes become real Unity objects called OvoShape.
- The ball's physics is a line-by-line port of the web code with the same numbers (gravity 1700, move 1400, max speed 850, bounce 0.5, friction 0.75 / 0.9, swap push 700). Where the web game looked at the colour of a pixel, Unity asks which shape is on top at that point.
- The ball, rings and exit use a small shader that shows the opposite colour of what is behind them, like the web version.
- Controls: A / D or Left / Right arrows move, any other key swaps, the mouse does nothing.

1. Install Unity

1. Go to https://unity.com/download, install Unity Hub and sign in with a free Unity ID (Personal licence).
2. Unity Hub -> Installs -> Install Editor. Pick Unity 6.6 (it looks like 6000.6.x). If it is not listed, use the Archive link there or https://unity.com/releases/editor/archive.
3. Tick Windows (or Mac) Build Support, and WebGL Build Support if you want a browser version. Wait for the install.

2. Create the project

1. Unity Hub -> Projects -> New project. Editor version 6000.6.x.
2. Choose the template Universal 2D (click Download template if needed). Name it Ovo and click Create project.

The window: Hierarchy (left, everything in the scene), Scene view (centre, where you move things), Game view (tab next to it, what the player sees), Inspector (right, settings of the selected thing), Project (bottom, your files), Console (tab next to Project, errors show in red).

3. Make Unity read the keyboard

1. Edit -> Project Settings -> Player -> Other Settings -> Active Input Handling. Choose Input System Package (New) (or Both). Click Yes to restart.
2. If it says the package is missing: Window -> Package Manager -> Unity Registry -> Input System -> Install.

4. Export the map from the web game

1. Double-click ovivo-level-exporter.html (open it in Chrome). It is the same game plus one button at the top left.
2. Wait 2-3 seconds, then click Download level.json. It lands in your Downloads folder (about 1 MB).

5. Put the files into Unity

1. In the Project panel: right-click Assets -> Create -> Folder. Make Scripts, Shaders and Editor. The folder must be named exactly Editor.
2. Drag these from your computer into Unity:
   - Into Scripts: OvoGame.cs, LevelSettings.cs, OvoShape.cs, OvoMesh.cs, HazardZone.cs, NoSwapZone.cs
   - Into Shaders: OvoSprite.shader
   - Into Editor: OvoLevelImporter.cs
3. Wait for the spinner at the bottom right to stop. Open the Console: there must be no red errors. If there are, copy the first one and send it to me.

6. Turn level.json into the map

1. Top menu: Ovo -> Import level.json as map. Choose the level.json you downloaded. It takes 10-30 seconds.

2. Unity creates Assets/Ovo with a material, a Decor mesh and Level1.prefab, and opens the prefab for you.
3. Look at the Hierarchy: Level1 contains Start, Checkpoint1-3, Gate, Map and Decor. Map holds the real shapes, named S0, S1, S2 and so on. In the Scene view you see the full map with green circles (start and checkpoints) and a red circle (gate). Turn on the Gizmos button if you do not.
4. Click any S object in the Scene view or the Hierarchy. In its Inspector you can see:
   - Gray: 0 = black, 255 = white (these shape the world and collide). Any other number is decoration only.
   - Hazard: tick it and touching the shape kills the ball (the spikes and thorns are already ticked).
   - Order: drawing order, bigger is on top.
   - Points: the corners. Is Line and Width turn it into a thick line.
5. You can already drag shapes with the Move tool (W key), delete them, or press Ctrl+D to copy them.
6. Click the back arrow at the top of the Hierarchy to leave prefab mode. Ctrl+S saves.

7. Add the game and press Play

1. Hierarchy -> right-click -> Create Empty. Name it Game. Add Component -> OvoGame.
2. Fill in OvoGame:
   - Levels: Size 1, then drag Assets/Ovo/Level1 into Element 0
   - Shader: drag OvoSprite from Shaders
   - Cam: drag Main Camera
3. Delete Global Light 2D if you see it. Do not put Level1 itself in the scene: the game creates it.
4. File -> Save As -> Main inside a new folder Scenes.
5. Set the Game view dropdown to Full HD (1920x1080), click the Game view once, then press Play. Press Play again to stop.

The first second after Play builds a lookup grid of the map, so expect a very short pause.

8. If something looks wrong

- Menu Ovo does not exist: a red error in the Console, or OvoLevelImporter.cs is not inside a folder named Editor.
- Import says the shader was not found: OvoSprite.shader is not imported yet, or has a compile error (select it and read the Inspector).
- Everything pink: the shader does not work. Send me the error.
- Nothing visible in Play: check the three OvoGame slots, and that Level1 is in the Levels list.
- The ball does not move: part 3 setting, or click the Game view once.
- Slow in the editor: turn Gizmos off in the Game view, and try a Build (part 9).
- Red Console errors: copy the first one and send it to me.

9. Make a build you can send to friends

1. File -> Build Profiles. Choose Windows (or macOS / Web). Add Open Scenes so Main is listed.
2. Click Build and pick an empty folder. Run the game from there.

10. Making more levels and restrictions

Copy Level 1 and edit it

1. In Project, select Assets/Ovo/Level1, press Ctrl+D, rename the copy Level2, double-click it to edit.
2. Delete the shapes you do not want (select them in the Hierarchy and press Delete). Move Start, the checkpoints and Gate where you want them.
3. Select the Game object and make OvoGame -> Levels bigger. Drag Level2 into the new slot. After the gate, press any key once the white circle has finished and the next level loads.

Add a new shape

1. In prefab mode right-click Map -> Create Empty. Add Component -> OvoShape.
2. Set Points (for example Size 4 with -1,-1 then 1,-1 then 1,1 then -1,1), Gray 255 for white or 0 for black, and Order above every other shape (use 6000 or more; the maximum is about 32000).

3. On the Mesh Renderer, set Materials -> Element 0 to OvoPlain. Right-click the OvoShape header -> Rebuild.
4. Easier: duplicate an existing shape, move it, and edit its points. You can scale and rotate polygons freely. Do not scale a Line shape (its collision width would no longer match).

A white shape on top of black makes cave. A black shape on top of white makes a wall, pillar or island. The ball lives in the colour where it stands, so check every route with Play.

Rules that keep a level playable

- Corridors at least 250 px (2.5 units) wide. The ball is 32 px across.
- Walls at least 80 px (0.8 units) thick so the ball can ride inside them to go up.
- Always give a way up: a thick black wall beside a shaft, then a way out at the top by swapping.

Restrictions you can use now

- Max Swaps on Level1 (LevelSettings): limits swaps. Small dots over the ball show how many are left. 0 = unlimited.
- Hazard tick on any shape: deadly.
- HazardZone: add a Polygon Collider 2D and the HazardZone script to an empty object. Deadly invisible area.
- NoSwapZone: same with the NoSwapZone script. Swapping is ignored inside it.

Ideas I can build for you: timers, one-way colour gates, moving spikes, switches that open walls, a locked colour per level, and a Scene-view tool for painting caves with the mouse.

11. Adding your soundtrack later

Select Game, Add Component -> Audio Source, drag your music file into AudioClip, tick Loop and Play On Awake.

12. What to ask me next

- Paste any red Console error and I will fix it.
- Tell me what to change, a new restriction or a new mechanic and I will update the scripts.